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Smarter Sole Survival: Will Neuropathic Patients at High Risk for Ulceration Use a Smart Insole-Based Foot Protection System?

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Abstract

Objective: This study examined adherence to alert-based cues for plantar pressure offloading in patients with diabetic foot disease.

Method and Design: Participants (n = 17) with in diabetic foot remission (history of neuropathic ulceration) were instructed to wear a smart insole system (the SurroSense Rx, Orpyx Medical Technologies Inc, Calgary, Canada) over a three-month period. This device is designed to cue offloading to manage unprotected sustained plantar pressures in an effort to prevent foot ulceration. A successful response to an alert was defined as pressure offloading, which occurred within 20 minutes of the alert onset. Patient adherence, defined as daily hours of device wear, was determined using sensor data and patient questionnaires. Changes in these parameters were assessed monthly.

Results: Patients demonstrating increased adherence over the course of the study received more alerts (0.82 ± 0.31 alerts/hour) than patients whose adherence did not improve (0.36 ± 0.46 alerts/hour, $P = .156$). By the final stages of the study, participants who had received at least one alert every two hours were more adherent with offloading than participants who received fewer alerts (52.5 ± 4.1% vs 24.7 ± 22.4%, $P = .043$). Furthermore, duration of time from alert generation to successful offloading was significantly greater in the group receiving fewer alerts. This was measured in the third month with an effect size Cohen's $d = 1.739$, $P = .043$.

Conclusion: The results suggest a minimum number of alerts (one every two hours) is required for patients with diabetic neuropathy to optimally respond to offloading cues from a smart insole system.

Keywords

offloading, diabetic foot ulcer, physical activity monitoring, wearable sensors, wound healing, adherence, Orpyx, SurroSense Rx

In recent years, insoles have been designed with sensors that monitor continuous plantar pressure and provide the user with alert-based feedback when plantar pressures are too high.[1-4] Pressure-sensitive insole systems typically consist of two primary components: (1) a pressure sensor-embedded insole, which converts plantar pressure into an electronic signal, and (2) a transmitter which triggers an alert in the form of audio, visual, or tactile feedback.[2] Pressure sensors are often placed at the toes, metatarsal heads, and heel, as these correspond to bony prominences where inherently high pressures are often observed.[1,3]

An increasing number of studies have begun to use feedback devices to help at-risk populations avoid negative outcomes.[5,6] One such study by Sanghan et al used an in-shoe dynamic pressure system to alert hemiplegics when the average plantar pressure between their feet differed by more than

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5%.[6] Another study by Berengueres et al used an insole equipped with a force sensor to detect when over-pronation occurred in patients with hallux valgus.[5] To date, however, these types of devices have not been used to reduce the risk of foot ulceration among patients with diabetes.

The use of novel smart insoles among diabetics would broaden the applicability of insole feedback devices to a much larger patient population. Diabetes, and the accompanying complications associated with this disease, has become a worldwide epidemic. According to data from the World Health Organization, the global prevalence of diabetes among adults was 6.4% in 2010, affecting 285 million adults worldwide. Between 2000 and 2030, the number of adults with diabetes is expected to increase by 50-70% in developing countries, and by 20% in developed countries.[7] The global prevalence of diabetes is expected to increase to 642 million people by 2040,[8] of which about 50% will develop peripheral neuropathy (PN).[9]

Diabetic foot ulcers (DFUs) often develop as a result of PN and its associated loss of protective sensation (LOPS) of the foot. It is considered a major public burden in both developed and developing countries, and affects approximately 15-25% of those with PN.[9-12] In 2012 alone, the total costs related to diabetes in the United States were $245 billion, a 41% increase since 2007.[13] Approximately one-third of all diabetes-related costs in the United States were spent on DFUs, with two-thirds of these costs incurred in the inpatient settings.[13] A previous report by our group in which over 1 million cases of DFUs that presented to emergency departments (EDs) in the United States from 2006 to 2010 were studied, suggested a national cost of $1.9 billion per year for ED treatment of DFUs alone and $8.78 billion per year in costs for inpatient care for these patients.[14] Arguably, the most significant consequence of DFUs is lower extremity amputation (LEA), complicating more than 10% of cases. It is estimated that approximately 70% of such amputations are preventable.[15] These data suggest an important gap in the effective management of DFUs, which could be addressed with better preventative measures.

The current standard of care for the prevention of foot ulcers includes the following basic elements: "screening, risk classification, regular foot care, protective shoes and insoles, and diabetic foot education."[16] Few researchers have demonstrated that these interventions are effective in the prevention of DFUs,[15] but several barriers still limit the effective implementation of these preventive care measures, particularly with the associated challenges of providing regular foot care, and poor patient adherence to the use of protective shoes and insoles among those with diabetes and LOPS. Physical barriers, such as obesity and limited joint mobility of the hip, knee, or ankle, may prevent the patient from properly positioning the foot for inspection. If patients are able to position their foot, they may have such poor vision that they cannot see well enough to accurately carry out the inspection. In a cohort of 115 diabetic subjects, it was found that 54% of patients lacked the requisite vision or joint mobility to accurately assess the dorsal and plantar surfaces of the feet.[17,18] In a similar study, Locking-Cusolito and colleagues reported that <3% of 232 diabetic patients undergoing hemodialysis had the ability to perform comprehensive self-care behaviors. Of these patients, only 75% of subjects had adequate vision, 60% had adequate joint mobility, and 55% had adequate flexibility to perform self-care.[17] In addition, patient vigilance for regular inspection of their feet decreases over time. DFUs that appear post diabetes diagnosis are directly correlated with an increasing lack of vigilance, given that neuropathy effectively neutralizes the natural "pain alarm" mechanism. Thus, a technology that allows one to regularly inspect the foot, while assisting in the identification of high-risk plantar pressure states is desperately needed and could have a significant impact on health care systems and patient quality of life worldwide.

Foot ulceration is multifactorial, but the most significant combination of factors has been shown to be elevated plantar pressures in a neuropathic insensate foot.[19] Foot deformities and limited joint mobility create regions of elevated plantar pressures and are associated with an increased risk of foot ulceration.[18,20-24] Abnormal weight-bearing pressures contribute to ulceration in two possible ways: by damaging the skin and underlying soft tissue, or by locally restricting blood flow.

At present, these sustained high plantar pressures are typically managed through passive methods, such as orthotics. Rocker shoes, medial arch supports, and custom insoles can help to reduce pressure at the forefoot by a reported 16% to 52%.[25] However, due to their passivity, the efficacy of these devices is limited to physically altering the architecture of the foot, which does not encourage users to adjust their behavior by offloading harmful pressures. In addition, the architecture of orthotics varies among manufacturers, and several studies have found no significant long-term benefits when orthotics were used.[26,27] The proactive and continuous-monitoring systems found within a smart insole system provide a potential alternative to the passive redistribution of plantar load achieved through custom footwear. An insole that can actively monitor plantar pressure, and provide alerts when these pressures are too high, may ultimately reduce the risk of foot ulceration and help patients avoid the damaging repercussions to which ulcerations often lead.

This study examined patient adherence to alert-based offloading with a pressure-sensitive insole system (the SurroSense Rx®, Orpyx Medical Technologies Inc, Calgary, Canada). The effectiveness of such insoles depends largely on the willingness of the participant to wear the device and offload pressures following cue-based feedback. For this reason, the focus of the current study was first to examine acceptability, practicality, and effectiveness of wearing a smart insole/smartwatch system designed to manage unprotected prolonged plantar pressure. In addition, we examined rate of successful response to an alert (a test of adherence and

efficacy) and changes in patient behavior (eg, adherence to footwear) over time and in response to frequency of hourly alerts (to examine potential alert fatigue and the effect of alert frequency on user engagement over time). In addition, we examined two contradictories hypotheses. The first hypothesis was that a high number of alerts per hour could cause user fatigue and leading to user disengagement. Similarly, our second hypothesis was that a low number of alerts per hour could yield the perception of device ineffectiveness, and thus, lead to user disengagement. Specifically, the study sought to determine whether the number of alerts triggered by users affected their adherence, successful offloading responses, and overall satisfaction with the device.

Methods

Subject Recruitment

This was a prospective cohort study to evaluate the effectiveness of a novel pressure-sensitive insole system (SurroSense Rx, Orpyx Medical Technologies Inc, Calgary, Canada) in reducing the risk of ulcer recurrence in diabetic patients with a LOPS. It included adult patients (21 years and older) who were ambulatory, had a history of diabetic peripheral neuropathy (DPN), and who had a history of plantar foot ulcers over the last 12 months. Exclusion criteria for this study included: any active plantar foot pathology (eg, DFUs, chronic Charcot arthropathy); participation (or planned participation) in an intervention research study during the last three months or during the study period; ankle brachial index (ABI) of 0.5 or less; or the presence of any of the following: active malignancy, immunosuppressive disease, cognitive deficit preventing adequate participation, history of major LEA (ie, above ankle), significant lower extremity edema, which may limit the patient from being fit with standard diabetic footwear, or any other issue that, at the discretion of the investigator, rendered the subject ineligible for participation. If subjects had noncompressible vessels (ABI > 1.2), toe pressures were measured to determine a toe brachial index (TBI). A TBI > 0.65 was required for enrollment. In addition, patients who could not be accommodated in standard diabetic shoes, or were unable to walk for a minimum of 20 meters with or without an assistive device, were excluded from the study. All subjects were recruited from Southern Arizona Limb Salvage Alliance (SALSA) at the University of Arizona Health System, USA, or affiliated and local clinics in Tucson, Arizona, USA. The study received local institutional review board approval from the University of Arizona. Each participant signed an informed consent prior to participating.

Device Used

Participants were provided with a pair of New Balance 928 Diabetic Walking shoes (MW928, New Balance, Boston, MA) and a smart insole system (the SurroSense Rx system, Orpyx Medical Technologies Inc, Calgary, AB). The smart insole system consisted of two pressure-sensing shoe insoles and a smartwatch display device. The insoles collected pressure information from the plantar surface of the foot, and wirelessly transmitted this information to the smartwatch worn on the participant's wrist (Figure 1). Each insole consisted of eight individual pressure sensors: three distributed across the metatarsal heads, two along the lateral plantar surface, one at the heel, one at the great toe, and one in the distribution of lateral toes.

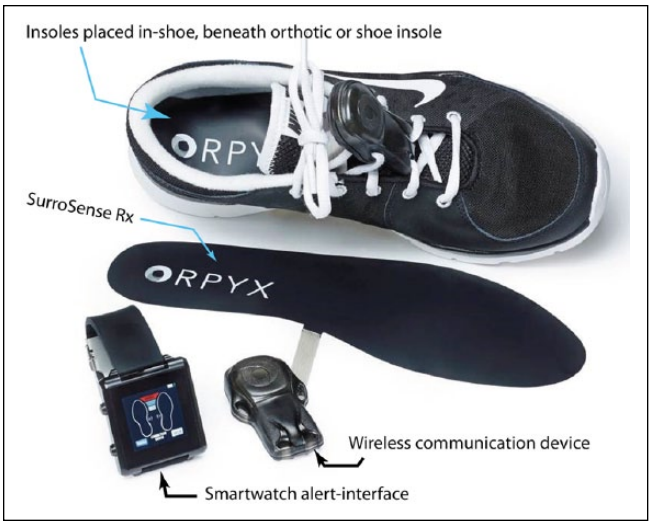


Figure 1. SurroSense Rx smart insole system (Orpyx Medical Technologies Inc, Calgary, Canada) for cuing plantar pressure offloading during activities of daily living. The insole contains eight pressure sensors located over regions considered to be highest risk for plantar foot ulcer development. The device was provided in standardized footwear (New Balance 928 shoes). The pictured smart watch is used as an interface to alert the patient when safe plantar pressure thresholds have been exceeded over time.

Participants were provided with an insole for each foot, unless they presented with an active ulcer, in which case they were provided with a single insole for the contralateral (nonulcerated) foot.

The device alerted the user when "safe" pressure and time thresholds were exceeded; these thresholds were based on the general clinical understanding of pressure ulcer formation, which indicates that a conservative threshold would be >35-50 mmHg for >15 minutes. If >95% of the measurements taken by a single sensor over a 15-minute scanning window exceed 35-50 mmHg, an alert was sent to the user via the smartwatch to guide them to appropriately offload that area (Figure 2).

Study Procedure

Figure 3 illustrates the study design. During the baseline appointment, participants received the smart insole system



Figure 2. Sample high pressure alerting screens from the SurroSense Rx smartwatch (Orpyx Medical Technologies Inc, Calgary, Canada). When an alert is triggered, the user receives auditory, vibratory and visual feedback from the watch. When the user acknowledges the alert, a map of each connected insole is shown, with the area of the high pressure alert being identified in red.

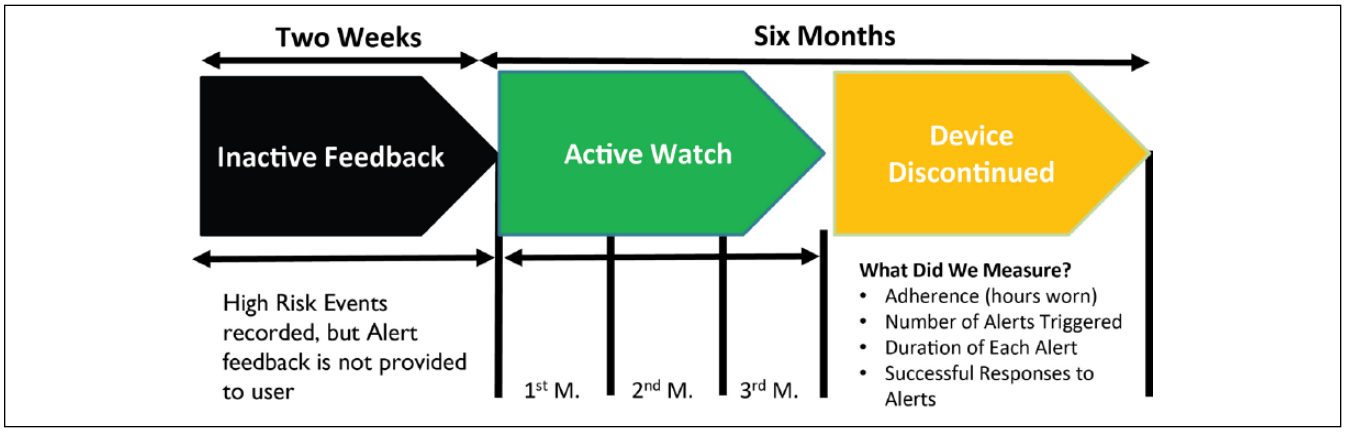


Figure 3. Study design: the study included three phases. During phase I (two weeks), participants wore the device, but the alerting function was rendered inactive. During phase II (three months), the alerting functionality was activated. During phase three (three months), usage of the device was discontinued. Phase II (active alert phase) was divided into three observation segments: month one, month two and month three. Changes in adherence, number of alerts triggered, duration of each alert prior to successful offloading, and rate of successful responses to alerts were analyzed for each observation segment.

with an inactivated (nonalerting) smartwatch. The inactive watch recorded pressure data, but did not provide alert-based feedback (auditory and vibrational) to the user. Participants wore the inactive device for two weeks to obtain baseline data on pressure distribution for each participant. After two weeks, the inactive smartwatch was replaced with an active (alerting) version. The active watch recorded all the same pressure information as the inactive version, but provided alert-based feedback in the form of a vibration and an audible beep. After the three-month active period, the device was removed from the patient; however, subjects were followed for an additional three months to screen for any potential development of foot ulcers when the device was not worn. The current study focused on the period following activation of the alert-based feedback (or Active Watch, Figure 3). To evaluate changes in adherence of device wear, the active period was divided in to three segments, based on monthly monitoring; first month (first segment), second month (second segment), and third month (third or last segment).

To determine each participant's willingness to wear the device, wear-time per day was monitored and termed "adherence." Adherence data were gathered objectively using data recorded by the sensors, and compared to subjective reports by the participants as determined by a questionnaire at the end of the study. The questionnaire asked participants to estimate their wear-time per day as 0-2 hours, 2-4 hours, 4-6 hours, 8-10 hours, or 10+ hours. Patients approximating their wear-time as 0-2 hours were categorized as self-reporting their adherence as 1 hour, those in the 2-4 hours were averaged at 3 hours, and so forth.

An alert ended only when a participant had successfully offloaded pressure on the specific plantar area. The percentage of alerts lasting fewer than 20 minutes served as a measure for the percentage of "successful" offloading of alerts. An "unsuccessfully" offloaded alert was defined as any alert that lasted longer than 20 minutes.

To evaluate the level of participants' physical activities, spontaneous daily physical activity was monitored using a

Table 1. Likert-Type Scale Survey Used in This Study to Evaluate Perception of Benefit, Acceptability, and Ease of Use (Revised Technology Acceptance Model [TAM]).

	Strongly agree	Somewhat agree	Neutral	Somewhat disagree	Strongly disagree
Learning to use the device is easy					
The system is comfortable to wear/use					
The system is easy to use					
The system provides me with valuable information regarding the health of my feet					
The system is effective in preventing blister/ infections/ulcers					
The device is a useful tool for diabetes					
The device should be worn by diabetes					
Using the system has been a pleasant experience					
Visiting the clinic on monthly basis was convenient					
I feel safe using the technology					
The products' overall performance in the treatment and prevention of diabetic foot problem is good.					
I would buy the insoles if they were commercially available					

validated and unobtrusive wearable sensor (PAMSys™, BioSensics LLC, MA, USA)[28] incorporated in a comfortable shirt (PAMShirt™) worn by participants for 48 hours at baseline. PAMSys allows quantification of physical activity by percentage of each main posture (sitting, standing, lying, and walking), total number of taken steps per day, number of unbroken walking bout, gait speed, longest unbroken walking bout, and number and duration of postural transition (sitting-to-standing) per day.[28-30] For the purpose of this study, only the number of steps per day was considered to evaluate the level of participants' daily physical activities.

User Experience

To gauge user satisfaction with the smart-insole system and perception of benefit, participants were asked to fill out a Likert-type scale survey, which was designed based on the technology acceptance model (TAM).[31] This survey assessed patient perception of device usefulness, ease of use, satisfaction and usability (Table 1). The Likert-type scale was designed to examine how strongly subjects agreed or disagreed with statements on a five-point scale with the following anchors: (1) strongly disagree, (2) somewhat disagree, (3) nature, (4) somewhat agree, (5) strongly agree. An average value of 4 or higher indicates successful achievement in the TAM.

Data Analysis

Pressure data were downloaded from each smartwatch and analyzed using an algorithm developed with MATLAB R2014b (Mathworks, Natick, MA). To identify possible predictors of adherence rates, participants were categorized based on whether they received more (HA: high alert group) or less (LA: low alert group) than one alert every two hours the device was worn. This cut-off was obtained by examining the number of daily alerts received in those participants who showed improved adherence over the course of the study. Results are expressed as mean ± standard deviation (SD). One-way analysis of variance (ANOVA) and Fisher's exact test (or chi-square χ^2 as appropriate) were used to examine between-group differences in descriptive data. Homogeneity of variance between the groups was confirmed with a Levene test, and a Welch's t-test was performed when the assumption of homogeneity of variance proved to be false. Spearman correlation coefficient was used to examine correlation between changes in adherence over time and number of alerts triggered per hour. Cohen's *d* effect size was used to estimate mean differences effect sizes between groups. Cohen's *d* values of 0.2, 0.5, and 0.8 were considered as small, medium, and large effect size, respectively.[32] All the statistical analyses were performed using SPSS (IBM, version 22, Chicago, IL), with a significance level of $P < .05$.

Results

Out of 17 participants who satisfied inclusion and exclusion criteria and were recruited for the purpose of this study, 12 completed the study and are included in this analysis (age, 62 ± 9.4 years; BMI, 33 ± 6.3 kg/m2). The reasons for not completing the study include hospitalization during inactive phase of the study (n = 1), one participant developed compression wound on dorsal foot during inactive phase of the study, technical problems (n = 2), and one participant had

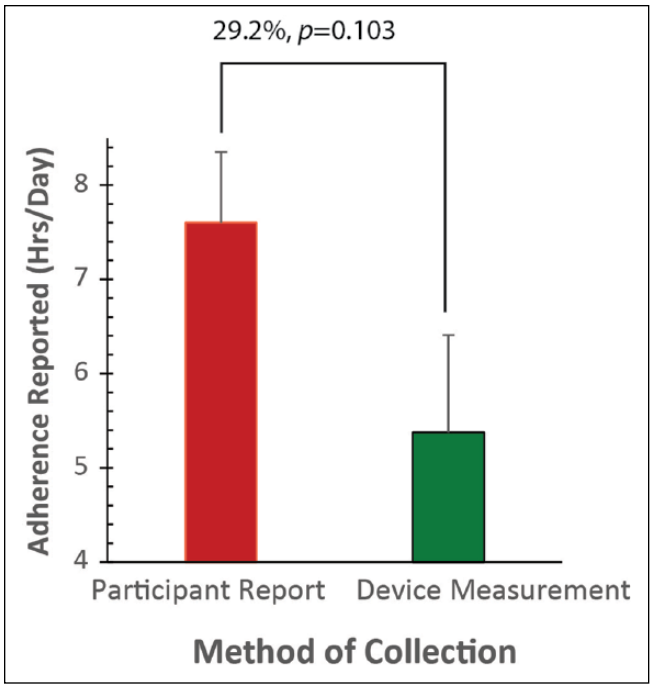


Figure 4. Wear time: average wear time of smart insoles per day, as reported by each participant, and as captured by the device. On average, participants tended to overreport their adherence by 29.2%.

their smartwatch stolen during the study, thus only partial adherence and alert data were available.

Participants wore the smart insoles for an average of 5.38 ± 3.43 hours per day, and received an average of 3.38 ± 3.81 alerts per day for the duration of the study. No adverse events were reported and no plantar ulcers were reported during the active phase of the study.

Adherence

Self-reported adherence (7.60 ± 2.50 hours/day) was higher among participants than the number of hours recorded by the smart insoles (5.38 ± 3.43 hours/day). Subjects over-reported their wear time by an average of 2.22 hours or 29.2% (P = .103; Figure 4), revealing a trend that patient reports of adherence are overestimated. However, this trend did not achieve a statistically significant level in our sample.

Among all participants, average adherence declined slightly from the first segment of the study (6.30 ± 4.58 hours) to the second (5.27 ± 3.28 hours, P = .50) and third (5.18 ± 4.24 hours, P = .48), but these changes were not found to be statistically significant in our sample. By exploring subject level behavior, we noticed that six subjects increased adherence between the first and third segments, while others decreased their adherence. Separating these participants into two groups, and comparing alerts per hour, revealed that those participants whose adherence improved trended toward receiving more alerts (0.82 ± 0.31 alerts/hour) than those whose adherence did not improve over time (0.36 ± 0.46 alerts/hour, P = .156) (Figure 5A). By fitting a linear curve for modeling change in adherence over time and average number of alerts triggered per hour worn per subject, we estimated that a cut off of 0.5 alerts per hour (or one alert per two hours) could be used as optimum cut off to separate between those who improved adherence over time compared to others (Figure 5B). The participants were then classified into groups: HA group, which included those who received at least one alert per two hour, and LA group, which included those who received less than one alert per two hours on average.

Table 2 summarizes the between-group comparisons between the HA and LA groups. None of the baseline demographic and clinical data including age, BMI, gender, and HBA1C were different between groups, but daily adherence was significantly higher in the HA group compared to the LA group with a large effect size (8.0 ± 1.8h in HA v. 3.6 ± 1.8h, Cohen's d = 1.512, P = .035). The number of daily steps was, on average, 6.4% higher in the HA group with a medium effect size (Cohen's d = 0.628), but did not achieve a statistically significant level in our sample. The effect sizes for other assessed demographic and clinical data were small for separation between groups.

Successful Responses

Interestingly, while the successful response rate in the HA group improved over time, it was reduced in the LA group (Figure 6A). During the first segment (1st month) of the alerting period, the HA group had similar rates of successful response to the LA group (41.4 ± 7.2% in HA v. 45.0 ± 16.9% in LA group, P = .699, Table 2). By the second segment (second month), the successful response rate in the LA group dropped to 25.6 ± 19.9%, while the successful response in the HA group rose to 44.1 ± 4.4%. The difference between groups did not achieve a statistically significant level during the second segment of the study (P = .110), despite the large effect size (Cohen's d = 1.283). During the last segment of the study, the successful response rate was further reduced in the LA group, and improved in the HA group (52.5 ± 4.1%), which was significantly higher than the LA group, with a large effect size (Cohen's d = 1.7126, P = .043).

Similar trends were observed by examining the duration of alerts that were not responded to (Table 2, Figure 6B). During the first segment of the study, the duration of unresponsiveness to alerts was almost the same between the two groups (13.5 ± 2.8 min in HA vs 19.6 ± 18.5 min in LA, P = .540). The duration of unresponsiveness to alerts became shorter over time in the HA group and reached 8.8 ± 1.7min during the last segment of the study. This duration was consistently longer in the LA group and reached on average 57.3 ± 39.4min, which is significantly longer than the HA group, and had a very large effect size (Cohen's d = 1.739, P = .043).

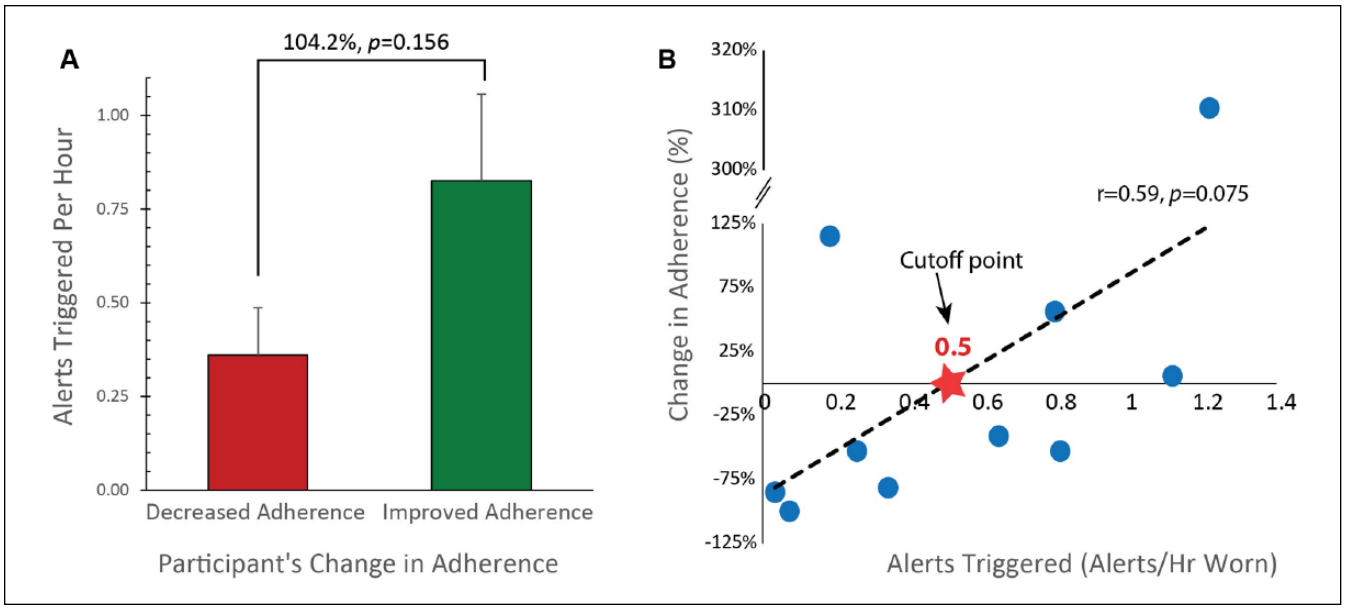


Figure 5. (A) Those who showed improved adherence over time had a 104.2% higher number of alerts per day compared with those who showed reduced adherence over time. (B) Those with at least one alert every two hours tended to show increased adherence over time. The star on the figure illustrates the location where the regression line cross x-axis (ie, zero change in adherence).

Table 2. Comparison Between Those Receiving a High Frequency of Alerts (an Average of at Least One Alert per Hour) and Those Receiving a Low Frequency of Alerts.

Parameter	Group	Mean ± SD	*P* value	Effect size, Cohen's *d*
Age, years	Low alert	59.8 ± 10.9	.501	0.483
	High alert	64.3 ± 7.4		
BMI, kg/m^2	Low alert	33.4 ± 6.5	.793	0.165
	High alert	32.3 ± 6.8		
Gender, % male	Low alert	50%	.548	—
	High alert	33%		
HbA1C (mmol/mol)	Low alert	8.2 ± 1.5	.253	0.304
	High alert	7.8 ± 1.1		
# of steps per day	Low alert	5695 ± 82	.486	0.628
	High alert	6059 ± 815		
Daily adherence, hours	**Low alert**	**3.6 ± 1.8**	**.035**	**1.512**
	High alert	**8.0 ± 3.7**		
Successful response rate (1st month), %	Low alert	45.0 ± 16.9	.699	0.277
	High alert	41.4 ± 7.2		
Successful response rate (2nd month), %	Low alert	25.6 ± 19.9	.110	1.283
	High alert	44.1 ± 4.4		
Successful response rate (3rd month), %	**Low alert**	**24.7 ± 22.4**	**.043**	**1.726**
	High alert	**52.5 ± 4.1**		
Unresponded alert duration (1st month), min	Low alert	19.6 ± 18.5	.540	0.461
	High alert	13.5 ± 2.8		
Unresponded alert duration (2nd month), min	Low alert	51.1 ± 55.4	.203	0.999
	High alert	11.9 ± 2.1		
Unresponded alert duration (3rd month), min	**Low alert**	**57.3 ± 39.4**	**.043**	**1.739**
	High alert	**8.8 ± 1.7**		

Bolded parameters indicate statistical significant level ($P < .050$). Successful response rate: indicates the percentage of high pressure alerts which are successfully offloaded by the subject. Unresponded alert duration: median duration of active alerts without successful response.

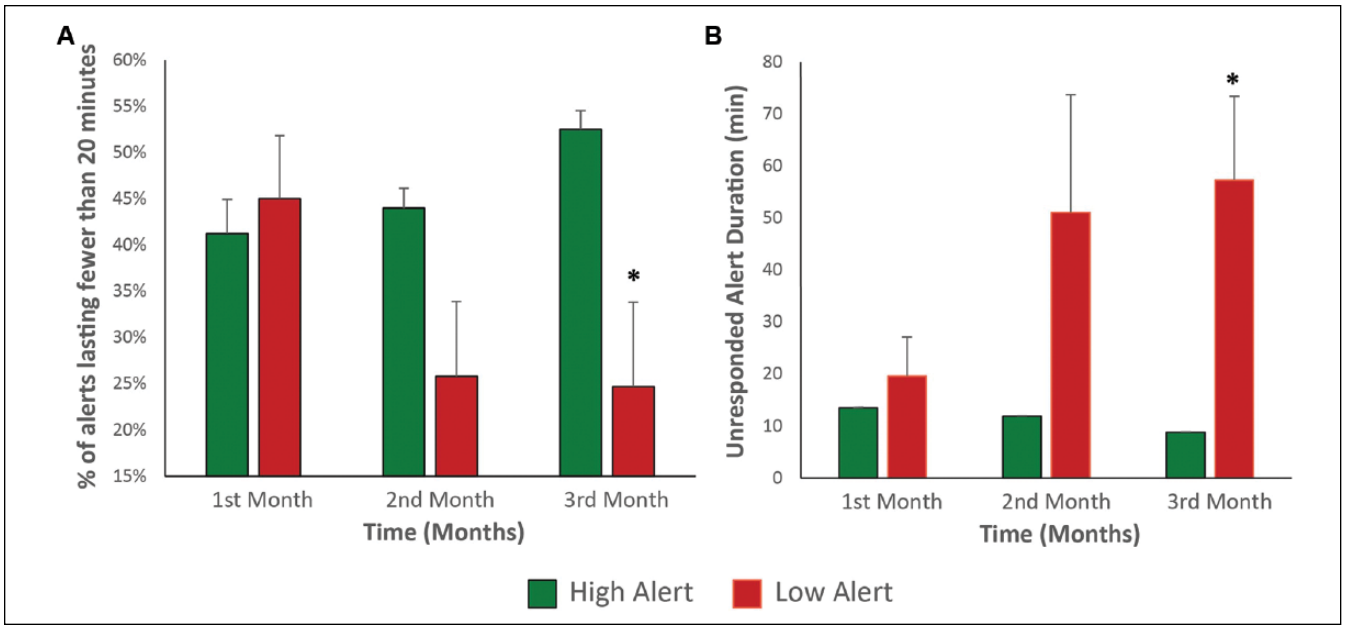


Figure 6. (A) Those in the high daily alert group (minimum one alert every two hours) showed an improvement in offloading compared to the low alert group in the third month of active device use. "Successful offloading of alerts" was defined as percentage of those alerts that were lasting fewer than 20 minutes. (B) Similarly, by the third month, those in the low alert group took longer to notice and/or successfully respond to alerts compared with those in the high alert group. *P value of less than .050.

User Experience

Overall, participants tended to agree that the device was useful for diabetics, was pleasant to use, and that the overall performance of the shoes was high (Figure 7). Participants were grouped based on the number of alerts they received (HA or LA group), and their subjective experiences were compared. Participants in both alert groups were satisfied with usage of the device, but HA users tended to rate the overall performance of the system higher than those in the LA group (4.6 ± 0.5 vs 4.0 ± 1.2), although the difference was not found to be statistically significant (P = .351). HA users more strongly agreed that wearing the device was a pleasant experience (5.0 ± 0.0 vs 4.4 ± 0.55) and was considered useful for diabetics (5.0 ± 0.0 vs 4.4 ± 0.55) (Figure 7B). Statistical significance could not be assessed because the HA group had no variance in the sample analyzed. When participants were asked if they would purchase the smart insole system, HA users (4.6 ± 0.5) responded more favorably than LA users (3.8 ± 1.3), although the results were not statistically significant in our sample (P = .240).

Discussion

This study examined patient adherence with a novel pressure-sensitive insole system (the SurroSense Rx). This mobile health device assesses plantar pressures over time, and provides the user with alert-based feedback when plantar pressure and time thresholds have been exceeded. The results of this study suggest that such technology is accepted by, and perceived to be beneficial to, diabetic patients with a high risk of DFUs. This was confirmed by both the TAM survey, as well as the objective monitoring of daily adherence regarding device wear time, duration of alerts, as well as frequency of "successful" offloading in response to alerts.

There are two dimensions to adherence in this study: (1) physical wearing of the device; and (2) responding to alerts in a timely manner. We tried to address both aspects in this study. The main goal of this study was to examine patient adherence to alert-based offloading with a pressure-sensing insole system. For this purpose, we first examined whether patients with DPN and a history of ulceration would agree to wear the studied system for sufficient periods of day to day wear, as defined by Chantelau and Haage.[33] We also studied whether they would respond to alerts in a timely manner (under 20 minutes). Second, we examined whether number of daily alerts could have an impact on daytime adherence to the device. We believe our definition of daytime adherence is acceptable for the purpose of our aims. However, adherence to the prescribed footwear could be biased by adherence to the device in question, as one system is fitted into the other and yet may be removed at the option of the patient. In other words, when device data are not measured, it was impossible to know whether the patient was not wearing their prescribed footwear, or whether they were wearing the inserts and no watch, or whether they had removed the device from their footwear. However, we believe this has no significant impact on our results, since the case of not wearing any component of the device could be interpreted as lack of adherence as

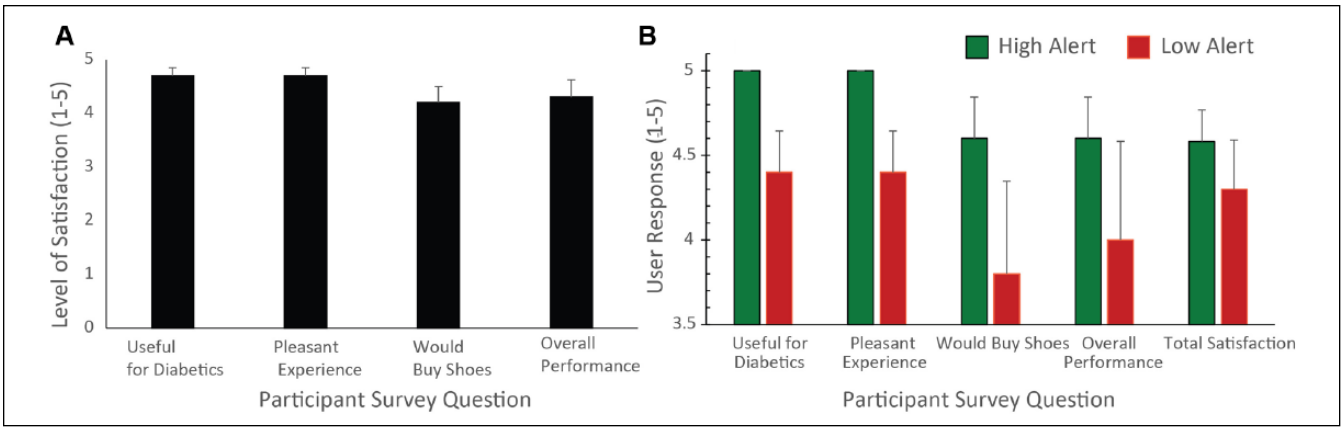


Figure 7. (A) Participants grades for perception of benefit, ease of use, and overall satisfaction. (B) Those who received a minimum of one alert per hour were more satisfied compared to those who received a lower frequency of alerts.

well. The device used in this study records the daily duration in which the footwear plus insole and smartwatch have been worn together, and we defined adherence with respect to wear as just that.

Interestingly, this study suggests that timely and frequent alerts about harmful foot loading activities with prolonged high pressure could enhance adherence to footwear overtime. This is aligned with other studies that suggest electronic reminder devices offer potential advantages to improve patients' adherence to interventions, medications, and therapies.[34-36] To the authors' knowledge, this is the first study that has demonstrated the effectiveness of in-shoe plantar pressure monitoring and a reminder device to enhance adherence to footwear in patients with diabetes and who are at a high risk of developing DFUs.

Footwear plays an essential role in foot health, particularly for those patients with diabetes and LOPS. Appropriate footwear is of the utmost importance in reducing harmful pressure on the bottom of the foot for the prevention and treatment of diabetic foot ulceration. However, clinical studies have demonstrated limited effectiveness of offloading footwear, and attributed this to noncompliant use by patients. In other words, prescribed footwear can only be effective in preventing DFUs if worn by the patient.[37] Several studies have suggested that high risk patients wear their prescribed footwear during only 15-28% of their total daily activities.[38-40] While education could play an important role in enhancing patient adherence to prescribed footwear,[41] recent studies suggest that without frequent education, the adherence could be diminished over time.[42] In this study, users in the LA group wore the device for an average of 3.6 hours, which is equivalent to about 30% of daytime. In comparison, patients who received at least one alert every two hours wore the device for an average of 8.0 hours per day, which is equivalent to 67% of daytime. Chantelau and Haage have demonstrated that patients who wore protective footwear for more than 60% of the day reduced the ulcer recurrence rate by more than 50%.[33] Therefore, it could be concluded that frequently alerting patients about harmful plantar pressures could be used as educational intervention to enhance adherence to prescribed footwear, and may significantly reduce the risk of developing recurrent ulcers.

Frequent reminders regarding plantar pressures could play an important role in increasing the awareness of high-risk diabetic patients about permanent risks, which can threaten their foot health. According to a Macfarlane and Jensen study, only 11% of patients with a history of foot ulcers viewed their condition as being worse than that of most others with diabetes mellitus.[43] Macfarlane and Jensen concluded that "diabetic patients with a history of diabetic foot complications do not recognize the severity of their condition, or falsely assume that most other diabetic patients are "just as bad off" as they are. Even patients who perceive their general diabetic condition or their diabetic foot condition to be worse or much worse than that of other diabetic patients are no more likely to be compliant in the use of diabetic footwear."[43] This may explain why those with a high daily frequency of alerts demonstrated improved adherence over time. In other words, patients who received alerts more frequently when harmful plantar pressures were identified may better perceive their feet as "at risk," which in turn may improve their adherence over time. This aligns with the survey performed, which found that patients who received more alerts showed a tendency to agree more strongly that wearing the device is useful for them than those who did not receive frequent daily alerts. It is important to note that this is true to a point, as after a threshold of high-alerting has been surpassed, patients may become frustrated and adherence could in turn decrease. Another study should be addressed to identify upper bound threshold for unacceptable daily frequency of alerts, which may frustrate and disengage the users.

Another unique aspect of this study is the objective monitoring of everyday adherence to footwear during a relatively long period of time (3 months), and exploring month over

month changes in adherence to footwear. To date, our understanding about footwear adherence is based on subjective methods, including questionnaires, face-to-face interviews, or diaries.[33,37] This study suggests that participants overestimate their adherence to footwear use on average by 29%. These results align with previous studies, which suggest that subjective methods are not accurate and reliable in the assessment of footwear or medication adherence among the diabetic population, and may lead to a response bias or to missing data.[37,43,44]

Few studies have objectively monitored diabetic patient adherence to footwear. These studies used either accelerometer-based sensors or in-shoe temperature to monitor adherence;[39,45] however, these studies monitored participants for only a few days, which may not be sufficient to evaluate changes in adherence to footwear over time. In a previous study done by our group, it was demonstrated that diabetic patients with active foot ulcers showed a decline in adherence to offloading after 4 weeks following the initial fitting of an offloading boot designed to treat DFUs.[42] The current study confirms this observation, and suggests that in the LA group, adherence to footwear sharply declines at the second month of monitoring and continues to decline in the third month. Interestingly, the adherence to footwear continues to improve among those patients who received frequent daily alerts. Another study should be addressed to examine whether using such alert-based devices could also enhance adherence to offloading among patients with active foot ulcers.

To the authors' knowledge, this is the first study to examine the acceptability and effectiveness of mobile health technology in the management of harmful plantar pressure loading among patients with diabetes. The SurroSense Rx smart insole system, used in the context of this study, is a novel mobile health device that monitors plantar pressure over time, and provides the user with alert-based feedback when plantar pressure and time thresholds have been exceeded. The feedback provided by the device is displayed to the user via a smartwatch, which identifies the area of high pressure and encourages the user to offload pressure from that region. Through continuous monitoring and biofeedback, this novel device applies smart insole technology to help reduce the risk of ulceration among patients with diabetes and PN. Unlike previous studies, in which only a very high magnitude of peak plantar pressures (above 200 KPa or 1500 mmHg) during walking,[46] which are often short duration (in range of few hundred milliseconds), were considered as harmful plantar loading, this technology focuses on prolonged unbroken plantar pressures exceeding 35-50 mmHg over a 15-minute moving window irrespective of the activity. Harmful plantar pressures can arise from any foot loading conditions, including standing or during locomotion when shoes are too tight or callus is present, which in turn may cause sustained, but not necessarily very high pressures (eg, above 200 KPa). Thus, the proposed intervention, using the SurroSense Rx, could be more effective than focusing only on walking, in particular among diabetic patients where the duration of standing has been reported to be three-fold longer than walking, but has been often neglected among people with diabetes with a risk of plantar ulcers.[47] In addition, our recent study[42] suggests that although number of taken steps is a predictor of speed of wound healing, when a multivariable model is used, only the duration of prolonged standing was a significant predictor of unsuccessful wound healing at 12 weeks. Thus, it stands to reason that prolonged unprotected standing and sitting (with prolonged foot loading) could put the patient with DPN at risk of ulceration and should be avoided. It is possible that a system alerting the wearer to both prolonged, in addition to short but high pressures carries additional value than a system alerting the user only to the former. In addition, the key technical benefit for such an approach is that, unlike measuring instantaneous peak pressure during walking, measuring sustained plantar pressures does not require a high sampling frequency and thus optimizes battery life while reducing the overall cost of the system. Another study should be addressed to examine the effectiveness of this method[48] in the prevention of plantar ulcer recurrence, in comparison with other methods which focus primarily on reducing peak pressures during walking, rather than prolonged and steady pressures during other foot loading activities.

Although the results of this study are promising, the sample size (n = 12) is too small to be conclusive. That being said, the sample size in this study is comparable with previous studies that have objectively examined adherence to footwear.[39,45] An additional study is merited to confirm the observations of this study in a larger sample size. In addition, the device used in this study was unable to collect data during periods where the device was not worn, thus we are unaware about true adherence to footwear during foot loading conditions (eg, walking or standing at home). However, when compared with percentage of daily use of footwear, the adherence to footwear was comparable to previous studies, as described above. This study was not powered sufficiently to examine the benefit of the proposed device in reducing the risk of foot ulcer recurrence. Meanwhile, the fact that no ulcers were developed while the device was active, and no adverse events were reported, is promising. In addition, the fact that adherence, which is known to be an important factor in reducing ulcer recurrence, improved in the HA group may indicate that the proposed alert-based device may reduce the risk of ulcer recurrence in these high risk populations in whom ulcer recurrence is estimated to be between 30-40% in the first year following initial ulceration.[12] A subsequent, sufficiently powered and randomized controlled study should be carried out to clinically validate the effectiveness of this innovative device in preventing the recurrence of plantar ulcers.

Conclusions

The results of this pilot study suggest that smart insoles, together with an alert-based feedback system notifying

patients of harmful plantar pressures, are perceived to be effective and acceptable by high risk diabetic patients, and could enhance adherence to prescribed footwear. Specifically, this study suggests that users who receive at least one alert per two hours could enhance adherence to footwear over time, respond better to alert-based feedback, and better perceive the benefit of such a technology-based intervention. In addition, a low level of daily alerts may discourage diabetic patients from using smart insoles. Further study is warranted to confirm the observations of this study in a larger sample size and over an ample period of follow-up.

Abbreviations

ABI, ankle brachial index; ANOVA, analysis of variance; BMI, body mass index; CI, confidential interval; DFU, diabetic foot ulcer; DPN, diabetic peripheral neuropathy; ED, emergency department; HA, high alert; LA, low alert; LEA, lower extremity amputation; LOPS, loss of protective sensation; PN, peripheral neuropathy; SALSA, Southern Arizona Limb Salvage Alliance; SD, standard deviation; TAM, technology acceptance model; TBI, toe brachial index.

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